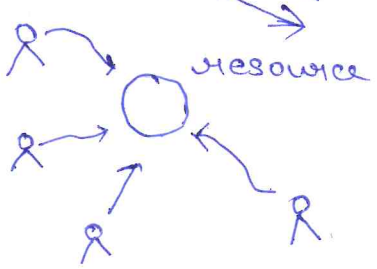
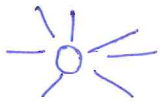
Design Idempotency Handler

* Idempotency vs Concurrency



BookMyShow



Idempotency help to take care of dublicate request.

* It enable clients to safely Retry an operation without worying about the side effect of operation can cause.

Operation

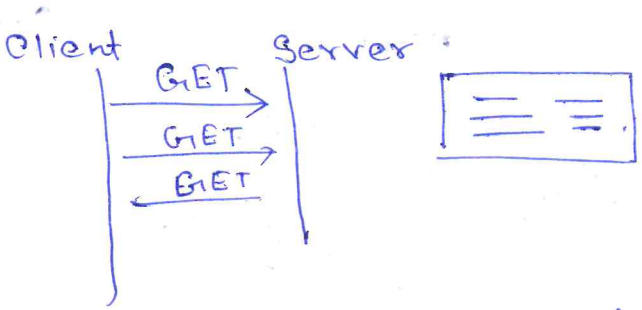
Adding an item into cart — POST → (Create Row)
POST (dublicat) →

Concept && Coding

① By default (GET, PUT, DELETE) are Idempotent in Nature.

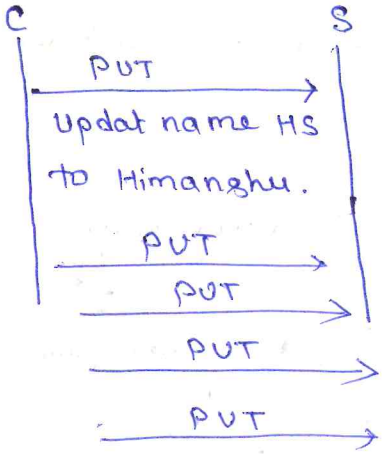
② POST is Not Idepotent & we have to make Idempotent

(i) GET



* It is not any side effect. Any time Client get request Server give data.

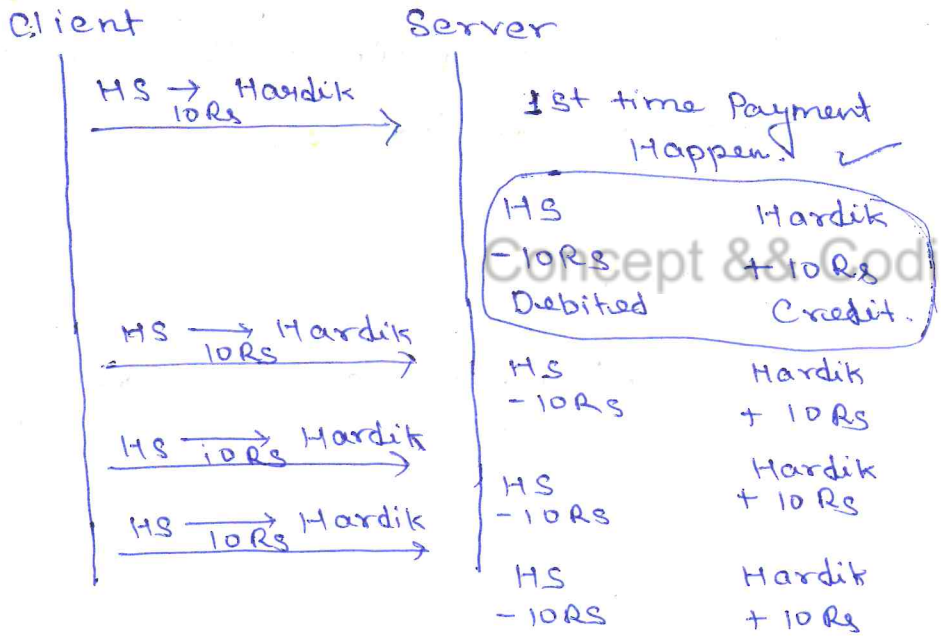
(ii) PUT



Changed
$$ – Himanshu

(iii) POST

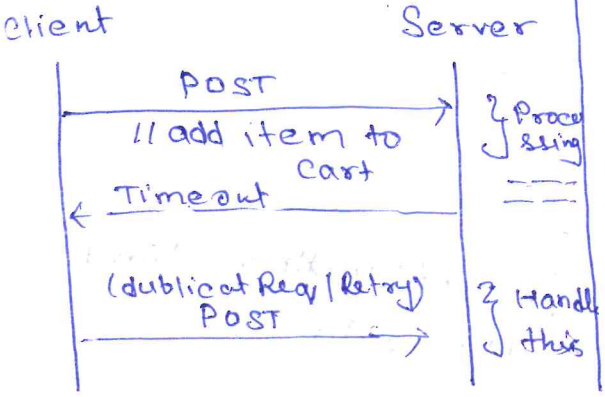
∵ Post is not idopotent in nanture.



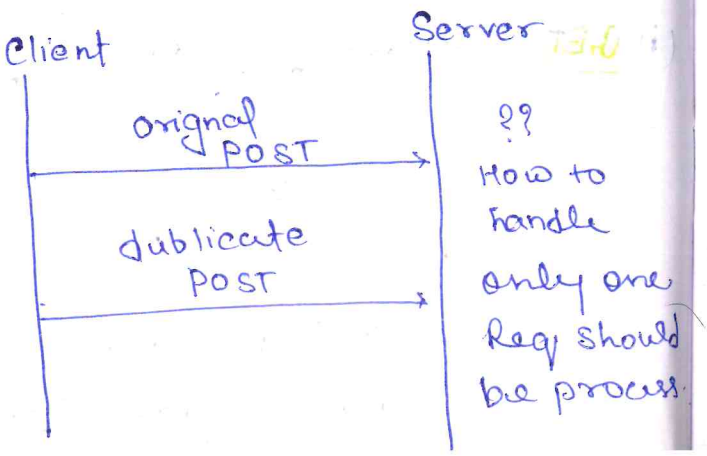
Concept && Coding

There are basically two type of Side effect : How dublicate can come.

① Sequential



② Parallel



Approach for Idempotency Handling

using Idempotency Key ?? (unique UUID) + append Operation

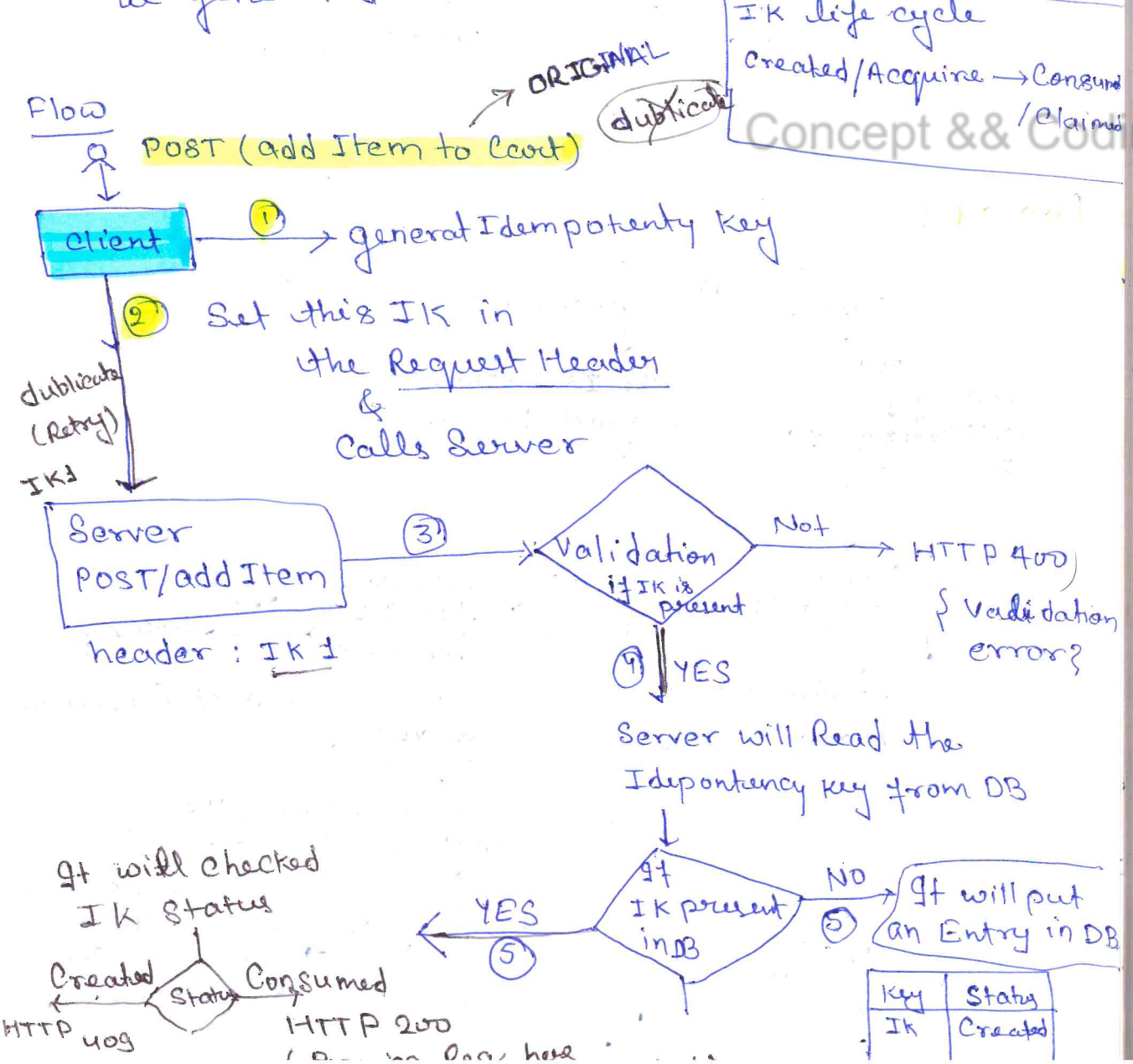
How ??

~~Aggreements~~ :

Aggrement with Clients :-

1. Client should generate the Idempotent key
2. For each different operation, new Idempotent key should be generated

IK life cycle
Created/Acquire → Consumed/Claimed



Key	Status
Ik	Created

Concept && Coding

If Idopontency Key is not present in DB So It will put the Entry in DB. after Entry in DB

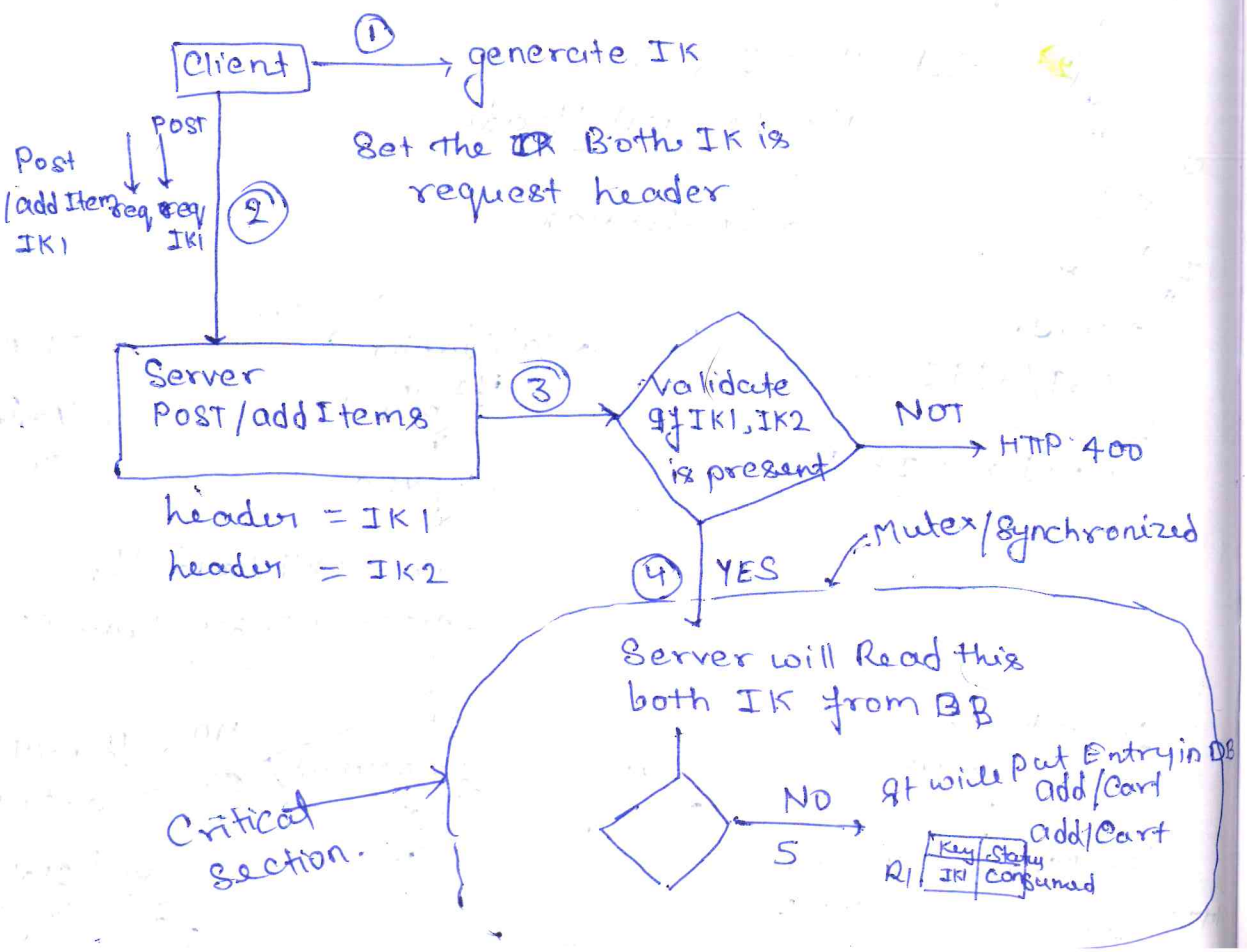
If It will put the Entry in DB

Key	Status
IK1	~~Created~~ Consumed

(6)

If call/operation is sucess It will Consume/Claimed the IK.

HTTP 201

Concept && Coding

Parallel :→